does ECMWF always use bulk richardson to get PBLH or does it use gradients of virtual tmemp. potenetial temp, sp humidity, relative humidity etc over land

Verified unified bulk Richardson method across all surfaces

ECMWF's IFS uses the bulk Richardson number method as its diagnostic for boundary-layer height — it doesn't switch to a separate gradient-based method over land versus ocean.

Specifics:

- The diagnostic follows the Vogelezang and Holtslag (1996) formulation of the bulk Richardson number, computed between the surface (or lowest model level) and successive model levels, incorporating both thermal (virtual potential temperature difference) and mechanical (wind shear) contributions, plus a surface-layer gustiness/turbulent velocity-scale term so the diagnostic still works in the well-mixed convective case where the local shear can vanish.
- PBLH is defined as the height where this bulk Ri first exceeds a critical value (documented IFS use is Ric ≈ 0.25, consistent with the Vogelezang-Holtslag scheme).
- This is purely a diagnostic — it's calculated as post-processing off the model state, and is separate from the actual turbulence closure that evolves the boundary layer. The prognostic turbulent mixing in IFS uses K-diffusion everywhere except unstable PBLs, where an Eddy-Diffusivity Mass-Flux (EDMF) scheme (Köhler et al., 2011) handles the non-local convective fluxes. That EDMF scheme internally uses its own entrainment/detrainment and updraft formulations, but the single number reported as "boundary layer height" (e.g., in ERA5 output) is still the bulk-Ri diagnostic, not a break in the vertical gradient of specific/relative humidity or potential temperature by itself.
- Simple vertical-gradient methods (looking for the level of maximum ∂θ/∂z, or the top of a specific-humidity gradient, or an inversion in relative humidity) are common in observational/radiosonde-based PBLH studies (and are used as reference/comparison methods in verification papers), but they are not what IFS itself uses to output PBLH. A 2019 evaluation of ECMWF IFS boundary-layer height against dropsonde data explicitly notes that although many methods exist to calculate PBL height, the bulk Richardson method is the one applied, which matches the model's own diagnostic definition. Wiley Online Library
- This treatment is the same over land and ocean; the IFS documentation doesn't describe a land-specific alternate diagnostic based on humidity or potential-temperature gradients. Land/ocean differences enter through the surface-layer exchange coefficients and roughness lengths (Louis et al., 1982 formulation), not through a change in the PBLH diagnostic method itself.

So: yes, it's bulk-Richardson-based throughout, not a gradient-based diagnostic, and this doesn't change between land and ocean surface types.

SO then the bulk richardson that I get using ERA5 137 level profiles over land, is quite a bit higher than they report in their output pblh

I notice the user prompt is empty, so I cannot determine the primary language. However, based on the th…

That's a very common finding, and there are several well-documented reasons a manually-computed bulk-Ri PBLH from ERA5 137-level profiles tends to run higher than the archived `blh` field, especially over land:

1. The official diagnostic isn't a "naive" bulk Ri — it has a built-in convective correction

ECMWF's internal PBLH diagnostic (following Vogelezang and Holtslag, 1996, building on Troen and Mahrt, 1986) doesn't just compute

$Ri\_b(z) = g \cdot z \cdot (\theta v(z) - \theta v,surf) / (\theta v0 \cdot (u^2 + v^2))$

using the raw lowest-level θv as the surface reference. In unstable (convective) conditions it adds:

- A **surface-layer gustiness/turbulent velocity term** to the denominator (something like $b \cdot u^{*2}$ added to u^2+v^2) so Ri doesn't blow up to unrealistically low values when actual wind shear is weak — very common over land at midday when winds near the surface are light and buoyancy-driven turbulence dominates.
- In many implementations, a **temperature excess** added to the surface parcel's virtual potential temperature to represent thermals, rather than using the bare lowest-model-level value.

If your own script uses the plain two-term bulk Ri without these correction terms, the behavior you get can go either direction depending on which effect dominates at a given profile — but the gustiness term is usually the big one over land: without it, weak near-surface winds make the denominator artificially small, and Ri can occasionally shoot past 0.25 lower in the profile in one layer, then not cross again until much higher up if there's a secondary near-surface stable layer (like a shallow superadiabatic layer right at the surface before the well-mixed layer proper) — so a naive calculation is prone to picking the *wrong* crossing point, often a higher one, especially if you're not careful about taking the *first* level from the surface where Ri exceeds critical, or if you're interpolating across large model-level spacing.

2. Level-crossing interpolation

With 137 levels, spacing near the surface over land is only a few tens of meters, but by the time you're a km up, spacing can be 200–400 m. If your code just steps level-by-level and doesn't linearly interpolate Ri between the last sub-critical and first super-critical level, you can overshoot by a full model layer, inflating PBLH by hundreds of meters at coarser resolution aloft.

3. Reference level mismatch

The official diagnostic typically uses the **lowest model level** (not 2 m temperature/humidity) as the surface reference. If you're using 2 m T/Td or skin temperature instead, especially over land at midday when there's a large gradient in the first few meters, you'll compute a different (often larger) $\Delta\theta v$ aloft needed to cross the threshold.

4. Forecast step vs analysis mismatch

The archived ERA5 `blh` is a **short-range forecast diagnostic** (from the 06/18 UTC forecast cycle), computed at the same instant as the analysis but not literally derived by ECMWF from the archived pressure/model-level fields you downloaded — it's produced online inside the model at full internal resolution and physics coupling. If you're recomputing Ri offline from the archived profile (which has been vertically discretized, possibly interpolated to pressure levels rather than kept on native model levels, and represents an average grid-box state), you're not exactly reproducing what the model's internal diagnostic saw at the physics timestep.

5. Land-specific noise

Land boundary layers, especially convective ones, tend to have noisier near-surface gradients (patchy surface heating, orography, soil moisture heterogeneity) than ocean columns. This makes a purely local Ri profile bounce around near threshold, so which crossing you pick (first exceedance vs. a more robust running criterion) can matter a lot — and the official scheme's extra terms exist partly to suppress that noise.

Practical suggestion: if you want your offline calculation to match ECMWF's number, use native model levels (not pressure-interpolated), the lowest model level as your surface point, linear interpolation to find the exact height where Ri = 0.25, and add a wind-gustiness term (something like +1 to 2 $(m/s)^2$ equivalent, or a scaled u^{*2} term) to the shear denominator rather than raw (u^2+v^2) alone. Comparing your PBLH against the official one on a scatterplot, stratified by stability (stable/neutral/convective) and by land cover, is a good diagnostic — mismatches concentrated in weak-wind convective daytime land cases would confirm it's the gustiness-term issue rather than a bug elsewhere in your code.

for the gustiness b * usqr, is there any dependency on surface temperature

Synthesized domain knowledge while acknowledging unverified internal model parameters

Not directly in the term's algebraic form, but yes indirectly, through how u* itself is diagnosed.

The term itself: In the Vogelezang & Holtslag (1996) formulation, the denominator of the bulk Ri is

$(u(z) - u_s)^2 + (v(z) - v_s)^2 + b \cdot u^{*2}$

The $b \cdot u^{*2}$ addition doesn't contain temperature explicitly — it's just a fixed multiple (b is an empirical constant, order 100 in the original Cabauw-tower fit) of the square of the surface friction velocity. So there's no separate T-dependent term bolted on next to it.

But u* itself is stability-dependent, and stability is set by the surface-to-air temperature difference:

u* comes out of the surface-layer similarity solution (Monin–Obukhov / Louis-type scheme in IFS), where

$u^* = \kappa \cdot U(z1) / [\ln(z1/z0) - \psi m(z1/L)]$

and the Obukhov length L is

$L = - u^{*3} \cdot \theta v / (\kappa \cdot g \cdot (w'\theta v')_s)$

$(w'\theta v')_s$, the surface buoyancy flux, is driven by the skin-temperature minus near-surface air-temperature difference (plus surface humidity flux). So:

- Over land at midday, strong surface heating (skin T well above the lowest-level air T) → large negative L → unstable ψm correction increases effective u* somewhat relative to the neutral log-law value for the same wind speed.
- At night, skin T below air T → stable, u* is reduced by the stability correction for the same wind speed.

So the chain is: surface (skin) temperature → sensible heat flux/buoyancy flux → Obukhov length → stability correction to the log-wind profile → u*. The $b \cdot u^{*2}$ gustiness term inherits that temperature dependence secondhand, through u*, not because temperature appears explicitly in the $b \cdot u^{*2}$ expression.

One more wrinkle worth flagging: some non-local schemes in this family (Troen–Mahrt 1986 type, used e.g. in YSU/KPP) instead build the unresolved-shear term from a *convective* velocity scale w* rather than (or in addition to) u*, and w* has a much more direct temperature dependence:

$w^* = [(g/\theta v) \cdot z_i \cdot (w'\theta v')_s]^{(1/3)}$

Since w* is a cube root of the surface buoyancy flux times a height scale, if the scheme you're replicating uses w^{*2} in place of (or added to) u^{*2} in the shear term, then yes, the temperature dependence becomes explicit and fairly strong (via the sensible heat flux). Whether ECMWF's operational implementation uses u*-only or blends in a w*-based term matters a lot for how sensitive your reconstruction is to skin-air temperature difference — it would be worth checking which exact velocity-scale term is coded in the current IFS Chapter 3 documentation (turbulent transport) rather than assuming it's identical to the original 1996 Cabauw fit, since ECMWF has tuned/modified the original formulation over successive cycles.

OK, so b=100 is a constant. But I need an estimate for ustar. I don't have any of these buoyancy terms. So can you distill all this into a simple form for me to use? I'll have skin temperature, u10,v10 and that is it. I don't even have T10 or T2

Weighed neutral drag coefficient approach against stability limitations

Fair enough — with only skin temperature and (u10, v10), you can't close the buoyancy/stability loop properly (that needs at least one temperature *difference*, and skin temperature alone doesn't give you one). Here's the honest, distilled version of what you can and can't do:

What you can compute: a neutral estimate of u*

$$u_* = \frac{\kappa\, U_{10}}{\ln(10/z_0)}$$

where:

- $U10 = \sqrt{(u10^2 + v10^2)}$
- $\kappa = 0.4$
- z0 = roughness length. For land you need this from somewhere — either ERA5's own forecast surface roughness field (`fsr`, a standard single-level ERA5 variable, essentially free to add to your download), or a fixed land-cover-based lookup (~0.03–0.1 m for open land, ~0.1–0.5 m for cropland/scattered trees, ~1+ m for forest/urban).

Then use $b \cdot u^{*2}$ with $b \approx 100$ as before.

What this misses: it's a *neutral* u*, with no stability correction. The real u* should be boosted somewhat in unstable conditions (convective, land daytime — exactly your problem case) and reduced in stable conditions, via Monin–Obukhov ψm(z/L). Since you have no air temperature at all, you can't form the surface buoyancy flux needed for L, so you literally cannot apply that correction — there's no way around this with the three variables you listed.

Practical implication of skipping it: the neutral u* will *underestimate* the true u* in unstable/convective conditions. That makes your gustiness term (b·u*²) *too small, which makes the denominator in your Ri formula too small, which makes Ri cross the critical threshold too low — i.e., this bias pushes your PBLH estimate down in exactly the convective land cases where you were seeing it come out too high. So it may partially offset the original problem, or it may not — it depends on which effect (missing gustiness term vs. neutral-u* underestimate of gustiness) dominates at a given point, and there's no clean way to predict that without testing.

My actual recommendation: grab `t2m` (2 m temperature) alongside skin temperature — it's a standard ERA5 single-level field, adds essentially no extra download cost, and gives you the one piece you're missing. With Ts (skin) and T2 you can form a proper bulk Richardson number for the surface layer itself and get a stability-corrected u* using the Louis (1979/1982) scheme — the same analytic (non-iterative) formulation ECMWF's own surface layer uses — rather than guessing with a neutral approximation. If you want, I can write out that Louis-scheme formula in a form you can drop straight into code once you have T2.

Yes can you write down that Lous scheme. Do I use over land and ocean, or on;y over land

Recalibrated formula using correct surface reference point

Good — that resolves it then. Let me give you the corrected, complete recipe using what you actually have (Ts, T2, u10, v10, plus the profile).

Rib numerator — use T2, not Ts:

$$Ri_b(z) = \frac{g\, z\, (\theta_v(z) - \theta_{v,2m})}{\theta_{v0}\, [(u(z) - u_{10})^2 + (v(z) - v_{10})^2 + b\, u_*^2]}$$

- $\theta_v(z)$ computed from your profile levels (T, q, p at each level)
- $\theta_{v,2m}$ computed from T2 (and 2 m specific/dewpoint humidity if you have it — if not, just use T2 as a temperature proxy for $\theta_{v,2m}$, accepting a small error from ignoring moisture)
- z measured from the surface (or from 2 m, being consistent)
- u10, v10 as the reference wind (matches the V&H formulation, which differences against near-surface wind, not zero)

Ts's job is separate — only for stability/u*:

Use Ts and T2 only to get Rib,surface (the Louis-scheme input) → stability correction → u*. That u* then only enters the $b \cdot u_*^2$ gustiness term in the denominator above. Ts should not appear anywhere in the main numerator.

Then:

- b = 100 (leave it there — you were right to not need to inflate it; the earlier symptom was a numerator problem, not a denominator one)
- z0 from `fsr` (land) for both the Louis calculation and the neutral CDN
- Critical Ri = 0.25, find the level (interpolating between bracketing levels) where Rib first crosses upward through 0.25 from below, scanning from the surface up

This should now behave properly: the previously "too high" bias driven by comparing air aloft against a hot skin temperature goes away, since T2 tracks the actual near-surface air rather than the radiometric surface skin value.

One thing worth checking once you rerun it: verify sign conventions and units are consistent (θ_v in K throughout, g=9.81, z in m) and that you're not double-subtracting the reference level twice (once in z, once implicitly if your profile's first level *is* 2 m — decide whether z is measured from 0 or from 2 m and be consistent between the wind and temperature terms). If it's still off after this fix, the next place I'd look is the u/v reference — some implementations difference against the *lowest model level* wind rather than u10/v10, and 10 m vs. lowest-model-level winds can differ by a couple m/s in stable conditions, which matters more than you'd think in the shear term.

better! not perfect but better. I will *not* have t2m for my retrieval, since I use a radiativeTransfer Algorithm that is coarser than the ERA5 137 levls.

should I just interpolate the profile (which is probably spaced at 125 m to 250 m resolution

Weighed interpolation pitfalls against using lowest available level directly

Don't interpolate — extrapolating a coarse profile down to 2 m will actively hurt you, not help, and here's why: the layer between your lowest usable level (~125–250 m) and the surface is exactly where the sharpest, most non-linear structure lives — the Monin-Obukhov constant-flux layer, where T follows a log-linear (not linear) profile in stable conditions, and where a strong nocturnal inversion or a shallow daytime superadiabatic layer can sit *entirely* below your resolution. Linearly (or spline-) interpolating your coarse levels down through that layer implicitly assumes the free-atmosphere lapse rate holds all the way to the ground, which is almost never true in the surface layer. You'd be manufacturing a fake T2 that's often further from reality than just using your lowest level directly.

Better approach: use your lowest available level as the reference, not 2 m.

Everything from before still applies — just substitute your lowest profile level (call it z1, T1, u1, v1) everywhere 2 m/10 m appeared:

- Surface Rib for the Louis stability correction: use (Ts, Tz1) and z1 instead of (Ts, T2) and 2 m.
- Rib numerator for the PBLH scan: reference θv(z1) instead of θv,2m.
- Wind reference in the shear denominator: u1, v1 instead of u10, v10.

This is conceptually identical to what NWP surface layer schemes already do — ECMWF's own lowest model level (~10 m) plays this same role, you're just doing it at coarser height. The math doesn't care what z1 is, as long as you're consistent.

The caveat you can't engineer around: Monin-Obukhov similarity is only strictly valid within the surface layer, roughly the bottom ~10% of the boundary layer depth. If z1 is 125–250 m and the actual PBL is deep (convective, >1500 m), you're probably fine — z1 is still within the constant-flux layer. But for shallow or stable PBLs (nocturnal land, often only 100–300 m deep), your lowest level can end up *at or above* the top of the surface layer, or even above the PBL top itself. In that regime, the whole basis for deriving u* from (Ts, Tz1) via MO similarity breaks down — there's no flux-layer relationship to exploit anymore, because z1 isn't really "in" the layer the theory describes.

There's no clean fix for that last case with the resolution you have — it's a genuine information limit, not a formula bug. Practically:

- Expect your worst residual errors to cluster in stable, shallow-PBL, low-wind land cases (nighttime), since that's where z1 most often exceeds the surface layer depth.
- Unstable/convective daytime cases should improve a lot, since the surface layer is typically deeper there and z1 stays within it.

- If you want a rough self-check, compare your diagnosed PBLH to z1 itself — if you're finding PBLH ≲ z1 or only marginally above it, that's a signal you're in the regime where the surface-layer assumption is shaky, and the number should be treated with more skepticism.